TSM 099

Homework 5

Common denominators and sums of fractions

Topics 21-23 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers. State every restriction from the original expression, including any value that makes a divisor zero. Cancel common factors, not terms.

Build common denominators

1. Find the LCM of 18 and 24. Then find the least common denominator of 1/(x squared - 1) and 1/(x squared + x). State excluded values.

2. Compute and reduce.

$$\frac{7}{12} - \frac{5}{18}$$

3. Compute and reduce.

$$-\frac{3}{8} + \frac{5}{12}$$

Add or subtract rational expressions

4. Combine, simplify, and state original restrictions.

$$\frac{2}{x} + \frac{3}{x+1}$$

5. Combine, simplify, and state original restrictions.

$$\frac{3}{x-2} - \frac{1}{x+2}$$

6. Combine, simplify, and state original restrictions.

$$\frac{x}{x-1} - \frac{2}{x-1}$$

7. Combine, simplify, and state original restrictions.

$$\frac{2}{x^2-x} + \frac{3}{x^2-1}$$

8. Combine, simplify, and state original restrictions.

$$\frac{4}{x^2-4} - \frac{1}{x^2+x-2}$$

Cancellation, signs, and checking

9. Combine and simplify. Retain original restrictions.

$$\frac{x}{x-3} - \frac{3}{x-3}$$

10. Simplify. Factor the denominator before choosing a common denominator.

$$\frac{1}{x-1} + \frac{2}{2-2x}$$

11. A student writes 1/x + 1/(x + 1) = 2/(2x + 1). Use x = 1 to disprove the claim, then derive the correct expression and state restrictions.